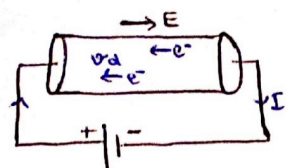


Electric Current & Current Density:



$$I = \frac{dq}{dt} = \frac{\Delta q}{\Delta t}$$

Flows from high potential to low potential.

Due to external electric field, the electrons inside conductor drift in opp. dirⁿ with $\vec{v}_d$ velocity.

Current density ($\vec{J}$)

$$\vec{J} = \frac{I}{A} \quad (\text{Amp/m}^2)$$

Area vector & dirⁿ of flow of current component should be same to each other.

Electric Mobility (μ)

Drift velocity can be defined in terms of electric mobility as

$$\vec{v}_d = \mu \vec{E}$$

$$\vec{v}_d = \frac{qE}{m} \cdot \tau$$

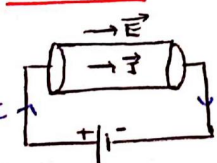
(τ → Relaxation time, time b/w two successive collisions of an e^-)

In terms of current density

$$\vec{J} = ne\vec{v}_d = \frac{ne^2 E \tau}{m} \quad (n \rightarrow \text{No. of free } e^- \text{ per unit volume})$$

$$\Rightarrow I = neA\vec{v}_d$$

Ohm's Law:



For a uniform conductor, Ohm's law gives the relation b/w its current density & ext. $\vec{E}$ at constant temp.

Ohm's Law (Differential Form)

$$\vec{J} = \sigma \vec{E} \quad \text{or} \quad \vec{E} = \frac{\vec{J}}{\sigma}$$

$$\sigma \rightarrow \text{Conductivity} = \frac{ne^2 \tau}{m} \quad \Rightarrow \quad \sigma = \frac{1}{\rho}$$

$$\rho \rightarrow \text{Resistivity} = \frac{m}{ne^2 \tau}$$

Normal Form of Ohm's law

$$\text{We know, } E = \frac{dV}{dl}$$

$$\Rightarrow \frac{dV}{dl} = \rho J \quad \int dV = \frac{\rho I}{A} \int dl$$

$$\frac{dV}{dl} = \rho \cdot \frac{I}{A} \quad \Rightarrow \quad \Delta V = \left(\frac{\rho l}{A} \right) I$$

$$(R = \frac{\rho l}{A}) \quad \Rightarrow \quad \Delta V = IR$$

Valid only for ohmic conductors where $\Delta V \propto I$

Current Electricity - 1

For very small temp. change $\Rightarrow$

$$\rho = \rho_0 (1 + \alpha \Delta T)$$

$$R = R_0 (1 + \alpha \Delta T)$$

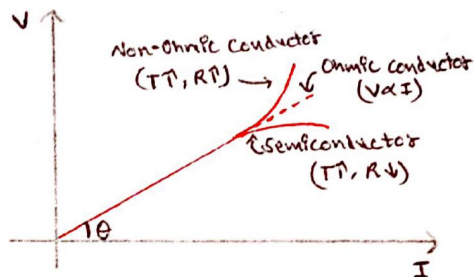
where, $\alpha \rightarrow$ Temp. coefficient

Verification of Ohm's Law $\Rightarrow$

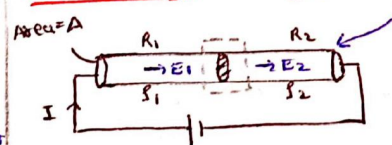
$\rightarrow$ For ohmic conductors, V vs I graph is a straight line.

$\rightarrow$ And slope of that graph gives Resistance.

$$\text{slope (tan } \theta) = \frac{\Delta V}{\Delta I} = R$$



Charge Accumulation at Junction of two Conductors: (Series combination)



$\rightarrow$ Both conductors have same cross section and same current passing through them.

$$I = \frac{V}{R_1 + R_2}$$

$$J = \frac{I}{A}$$

$\rightarrow$ Due to diff. resistivity (ρ), electric fields on either sides of the junction would be different.

$\rightarrow$ Which means ϕ_{electric} entering & exiting junction is not same, so some charge (q_s) will be left.

By Gauss Law $\Rightarrow$

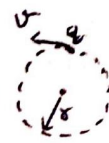
$$E_2 A - E_1 A = \frac{q_s}{\epsilon_0}$$

$$\rho_2 I - \rho_1 I = \frac{q_s}{\epsilon_0} \quad \left[\begin{array}{l} \rho = \frac{E}{J} \text{ \& } \\ E \cdot A = \rho I \end{array} \right]$$

$$\Rightarrow q_s = \epsilon_0 (\rho_2 - \rho_1) I$$

Circular Current:

q charge moving in a circle of r radius with const. velocity v .

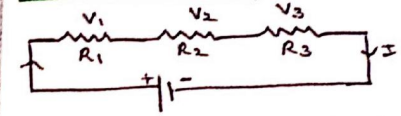


$$i = \frac{q}{T} = \frac{q\omega}{2\pi r}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}$$

Combination of Resistances:

Series Combination $\Rightarrow$



(i) Current across all resistances is same.

$$I = \frac{\Delta V}{R_{eq}}$$

(ii) Potential across each resistance is different and directly proportional to its resistance.

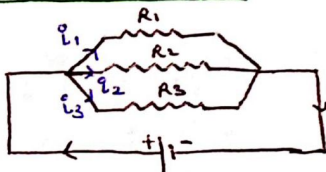
i.e. (V divides in Ratio of R)

$$(iii) R_{eq} = R_1 + R_2 + R_3$$

(iv) If N equal Resistances are connected in series, then

$$R_{eq} = NR \quad \text{and} \quad V_{\text{each}} = \frac{V}{N}$$

Parallel Combination $\Rightarrow$



(i) Current across all resistances is different, and they divide inversely to the value of Resistance.

i.e. (Jaha kam Resistance, waha zyada current flow hogi)

(ii) Potential across each resistance is same.

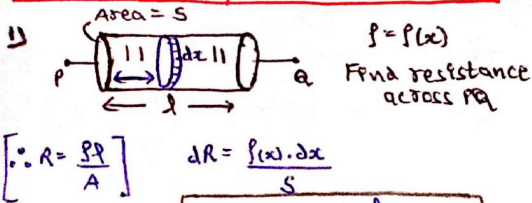
$$V = I R_{eq}$$

$$(iii) \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

(iv) If N equal Resistances are connected in parallel, then

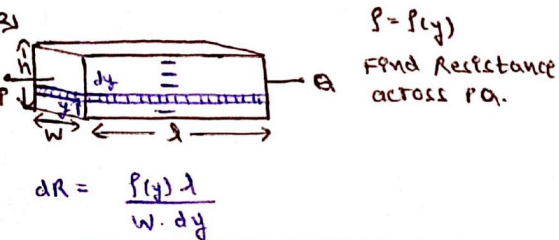
$$R_{eq} = \frac{R}{N} \quad \text{and} \quad I_{\text{each}} = \frac{I}{N}$$

Resistance with parameter variation:



$$\Rightarrow R_{\text{Total}} = \int dR = \int_0^l \frac{\rho(x) \cdot dx}{S}$$

→ Every element considered to be connected in series due to geometry.

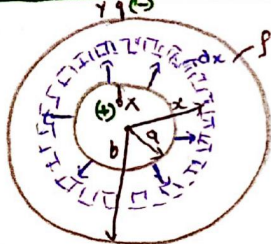


$$\text{So, } (R_{\text{Total}})^{-1} = \frac{1}{\int dR} = \int_0^l \frac{w \cdot dy}{\rho(y) \cdot l}$$

→ Every element considered to be connected in parallel.

Resistance for Radial Conduction:

Concentric Spherical shell:



Resistance,

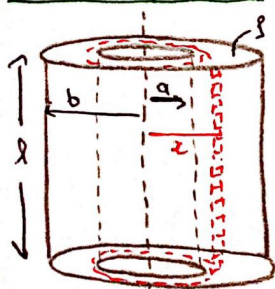
$$dR = \frac{\rho \cdot dx}{4\pi x^2}$$

$$R_{xy} = \int dR = \int_a^b \frac{\rho}{4\pi x^2} \cdot dx$$

$$R_{xy} = \frac{\rho}{4\pi} \left[-\frac{1}{x} \right]_a^b$$

$$\Rightarrow R_{xy} = \frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$$

Concentric Cylinders:



$$dA = 2\pi x \cdot l$$

$$\text{So, } dR = \frac{\rho \cdot dx}{2\pi x \cdot l}$$

$$R_{\text{Total}} = \frac{\rho}{2\pi l} \int_a^b \frac{dx}{x}$$

$$\Rightarrow R_{\text{Total}} = \frac{\rho}{2\pi l} \ln\left(\frac{b}{a}\right)$$

Equivalent Temp. Coefficient of combination of Resistances:

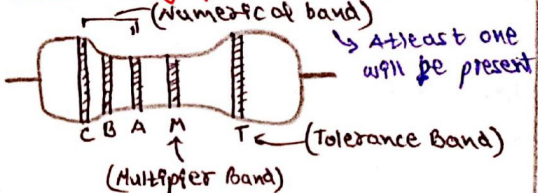
For a series or parallel combination of two resistances →



(For both cases)

$$x_{eq} = \frac{x_1 + x_2}{2}$$

Colour Coding of Resistances:



→ Always start reading bands from Tolerance band.

→ Sometimes, colour of Tolerance band can also be used to signify multiplier band.

Colour Code

Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Grey	8
White	9
Gold	0.1 - 5
Silver	0.01 - 10
None	20

① Always write tolerance by (+)

② In case of multiplier band, raise the power of 10 to the no. corresponding to its colour.

Eg: Green(M) → 10⁵
Black(M) → 10²

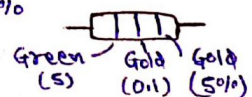
Pneumonic →

BB ROY of Great Britain
(Black) (Red) (Orange) (Yellow) (Green) (Blue)

Violated Law of Grey White
(Violet) (Grey) (White)

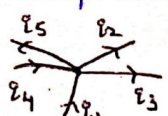
Eg: Find colour coding of $R = 0.5 \pm 5\%$

solⁿ $R = 5 \times 0.1 \pm 5\%$



Kirchoff's Current Law:

→ Electric current entering and exiting through a junction is always same.



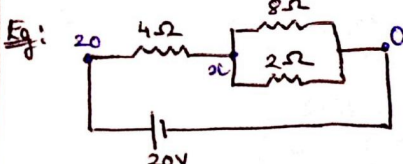
$$I_2 + I_3 + I_5 - I_1 - I_4 = 0$$

$$\Rightarrow I_1 + I_4 = I_2 + I_3 + I_5$$

→ It is extensively used to solve ckt's by finding unknown point's potential

→ Just make sum of current flowing in all branches connected to unknown point, equal to zero.

$$\text{Use, } [I = \frac{\Delta V}{R}]$$



For x

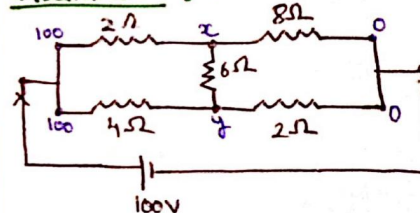
$$\frac{x-20}{4} + \frac{x-0}{8} + \frac{x-0}{2} = 0$$

$$\Rightarrow \frac{2x-40+x+4x}{8} = 0$$

$$\Rightarrow 7x-40 \Rightarrow x = 40/7 \text{ V}$$

Wheatstone Bridge:

Unbalanced →



For x

$$\frac{x-100}{2} + \frac{x}{8} + \frac{x-0}{6} = 0$$

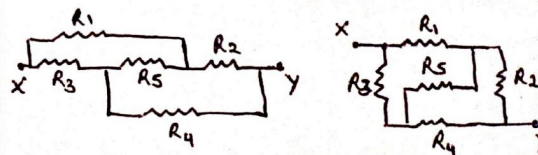
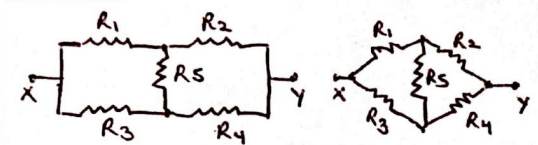
For y

$$\frac{y-100}{4} + \frac{y}{2} + \frac{y-x}{6} = 0$$

Solve both 1st eqn's & find x & y.

→ Solve further for required unknown parameter acc. to ques.

Balanced →

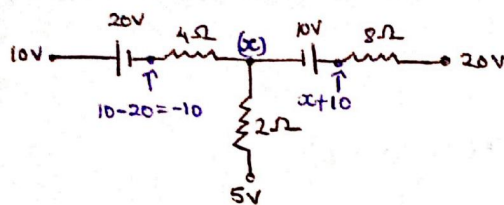


→ All are different representation of same circuit.

$$\frac{R_1}{R_3} = \frac{R_2}{R_4}$$

No current flows through R5 & it can be removed

Analysis of Circuit Branches:



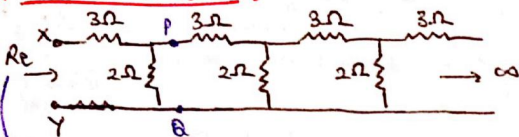
Here we use KCL for x

$$\frac{x - (-10)}{4} + \frac{x - (x+10) - 20}{8} + \frac{x - 5}{2} = 0$$

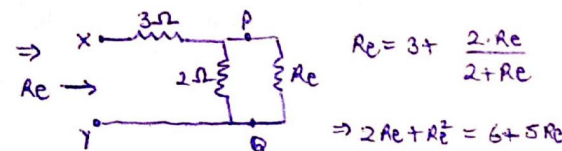
$$\Rightarrow \frac{2x + 20 + x - 10 + 4x - 20}{8} = 0$$

$$7x - 10 = 0 \Rightarrow \boxed{x = \frac{10}{7} \text{ V}}$$

Ladder Network:



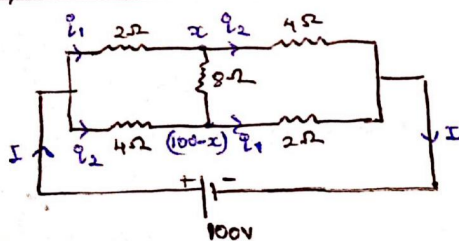
Equivalent Resistance of the entire Ladder Network.



Solve quad. and find the value of R_e

Symmetry in Circuits:

1) Symmetric Unbalanced Wheatstone Bridge



KCL eqn for x

$$\frac{x}{4} + \frac{2x - 100}{8} + \frac{x - 100}{2} = 0$$

$$8x - 500 = 0 \Rightarrow \boxed{x = 62.5 \text{ V}}$$

$$\text{So, } R_{eq} = \frac{100}{I} = \frac{100}{\frac{x}{4} + \frac{100-x}{2}}$$

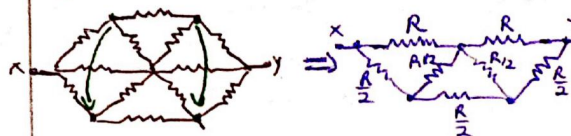
$$\Rightarrow \boxed{R_{eq} = \frac{800}{2.75} = \frac{32}{11} \Omega} \quad \text{Ans}$$

2) Point Folding Symmetry

If ckt is symmetrical about its terminal, then common points can be

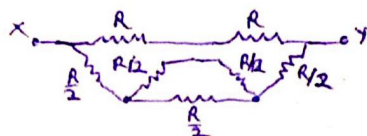
Current Electricity-2

joined together and overlapping resistances get connected in parallel. (All Resistances are same)

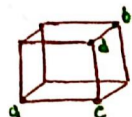


3) Current outgoing-Entering symmetry

Also, in above folded circuit, by this symmetry



Cube Resistors

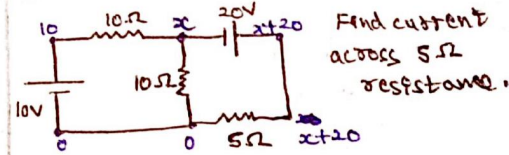


$$(R_{eq})_{ab} = \frac{5R}{6} \text{ (Body Diagonal)}$$

$$(R_{eq})_{ac} = \frac{7R}{12} \text{ (Edge)}$$

$$(R_{eq})_{ad} = \frac{3R}{4} \text{ (Face diagonal)}$$

Circuits with two or more Batteries:



Soln Applying KCL for x

$$\frac{x-10}{10} + \frac{x}{10} + \frac{x+20}{5} = 0$$

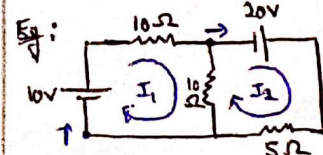
$$\Rightarrow 4x + 30 = 0 \Rightarrow \boxed{x = -7.5 \text{ V}}$$

$$\boxed{I_{5\Omega} = \frac{20 - (-7.5)}{5} = 2.5 \text{ A}} \quad \text{Ans}$$

Kirchhoff's Voltage Law (KVL):

→ Sum of all the potential drops in a loop is equal to the emf of battery in that loop.

$$\sum_{i=1}^n I_i R_i = \sum_{j=1}^m E_j$$



KVL eqn for I1

$$10 - 10I_1 - 10(I_1 - I_2) = 0 \quad \text{--- (1)}$$

KVL eqn for I2

$$+20 - 5I_2 - 10(I_2 - I_1) = 0 \quad \text{--- (2)}$$

From (1) & (2) →

$$\boxed{I_1 = 1.5 \text{ A}}$$

$$\boxed{I_2 = 2.5 \text{ A}}$$

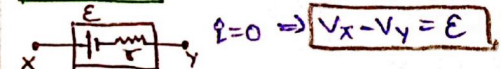
NOTE: Ifs loop ke lye KVL eqn krkh dene note hai, sign uska current +ve lye hai, baki sabka -ve.

EMF and cell grouping:

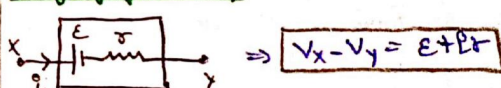
→ In any region, potential difference exist due to non-electrostatic reasons, where no current is flowing, then that pot. diff. is called 'Electromotive Force (EMF)'

→ In case of a battery, potential difference across its terminals in an open circuit (no current flowing) is 'emf (E)'

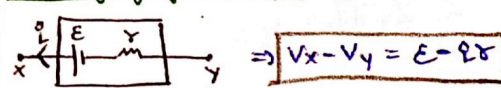
Open circuit →



Charging of battery →



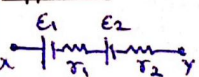
Discharging of Battery →



Cell Grouping

Series

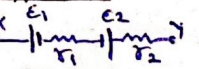
Supportive



$$\boxed{E_{eq} = E_1 + E_2}$$

$$\boxed{r_{eq} = r_1 + r_2}$$

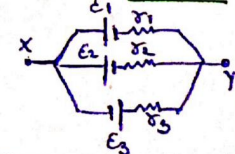
Opposite



$$\boxed{E_{eq} = E_1 - E_2}$$

$$\boxed{r_{eq} = r_1 + r_2}$$

Parallel

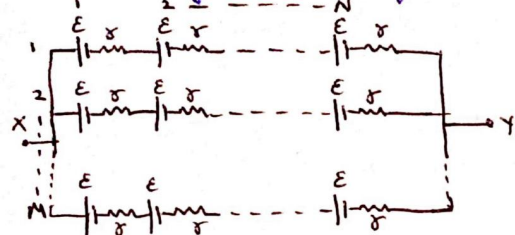


$$\boxed{E_{eq} = \frac{E_1}{r_1} + \frac{E_2}{r_2} - \frac{E_3}{r_3}} \quad \text{over } V_{r1} + V_{r2} + V_{r3}$$

$$\boxed{\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}}$$

Battery Grid :

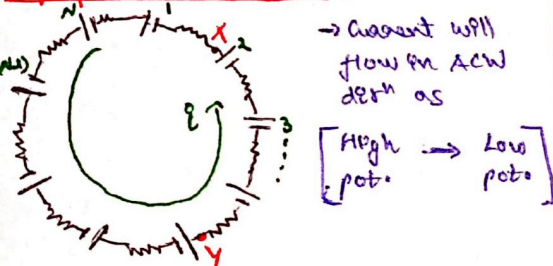
- N same batteries of $\mathcal{E}$ in series.
- M rows of such arrangement.



$$\mathcal{E}_{eq} = N\mathcal{E}$$

$$r_{eq} = \frac{Nr}{M}$$

Loop of Identical batteries :



For loop

$$\mathcal{E}_{eq} = Nr$$

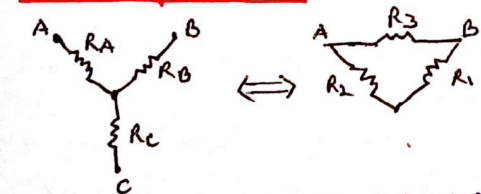
$$\mathcal{E}_{eq} = N\mathcal{E}$$

$$\mathcal{E}_{loop} = \frac{\mathcal{E}_{eq}}{r_{eq}} = \frac{\mathcal{E}}{r}$$

→ Also, the potential difference b/w any two random points on the loop is zero.

$V_x - V_y = 0$
⇒ $V_x = V_y$ Potential at all points on the loop is same, yet current is flowing.

Δ -Y Transformation :



$$R_A = \frac{R_2 R_3}{R_1 + R_2 + R_3}$$

$$R_B = \frac{R_1 R_3}{R_1 + R_2 + R_3}$$

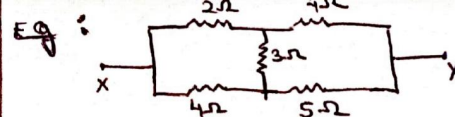
$$R_C = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

$$R_A = \frac{R_2 R_3 + R_2 R_1 + R_3 R_1}{R_2}$$

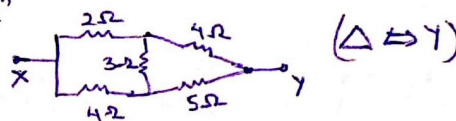
$$R_B = \frac{R_1 R_3 + R_2 R_1 + R_3 R_1}{R_3}$$

$$R_C = \frac{R_1 R_2 + R_2 R_1 + R_3 R_1}{R_1}$$

→ Both the forms are interconvertible and their resistances can be given by above formulas.



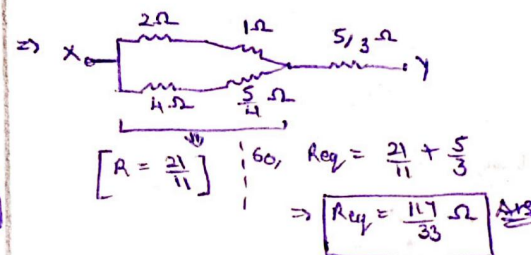
Solⁿ



$$R_A = \frac{3 \cdot 4}{12} \Rightarrow 1 \Omega$$

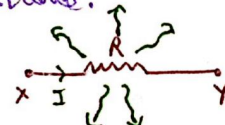
$$R_B = \frac{3 \cdot 5}{12} \Rightarrow \frac{5}{4} \Omega$$

$$R_C = \frac{4 \cdot 5}{12} \Rightarrow \frac{5}{3} \Omega$$



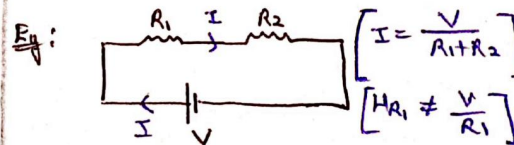
Thermal Effects of Current :

→ When current passes through a resistor, then some energy is lost in the form of heat dissipated through the resistor.



$$\text{Rate of Heating / Thermal Power} (H) = I^2 R = \frac{V_{xy}^2}{R}$$

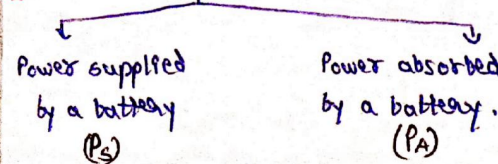
NOTE: The potential difference for heat calculation should be the one which is across the resistance and not that of battery (in most cases).



$$H_{R1} = I^2 R_1 \Rightarrow \frac{V^2}{(R_1 + R_2)^2} \cdot R_1$$

$$H_{R2} = I^2 R_2 = \frac{V^2}{(R_1 + R_2)^2} \cdot R_2$$

Battery Power :



Ideal Battery :

$$\text{Power supplied by battery } (P_s) = EI$$

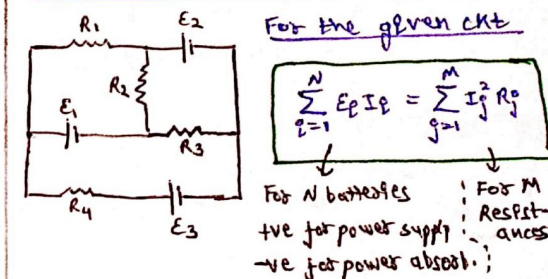
$$\text{Power absorbed by battery } (P_a) = EI$$

Non-Ideal Battery :

$$\text{Power supplied by battery } (P_s) = EI - I^2 r$$

$$\text{Power absorbed by battery } (P_a) = EI + I^2 r$$

Energy Conservation in an Electric Ckt :



Also, above powers are also equal to total Thermal Power (P_m) of the ckt.

Brightness of Bulbs in Ckt :

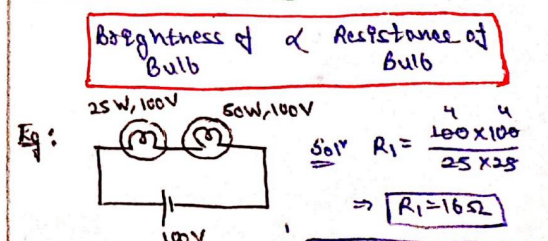
→ Rating of a bulb is given by (Power / Voltage)

→ By this Rating, we can find resistance of the bulb by

$$R = \frac{V^2}{P}$$

→ Bulb across which power is more, will be more brighter.

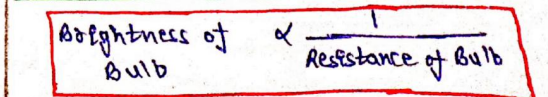
Series Combination :

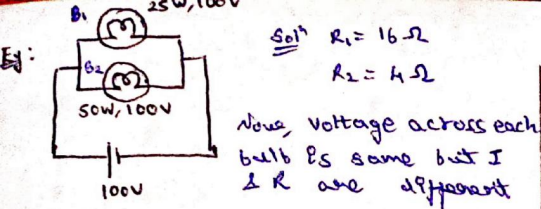


Since bulbs are in series, so same I flow through both.

$$P \propto R \quad (I, \mathcal{E} = \text{const}) \Rightarrow \text{Bulb with higher } R \text{ is brighter}$$

Parallel Combination :



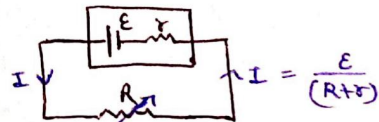


$$\Rightarrow \left[P = \frac{V^2}{R} \right] \Rightarrow P \propto \frac{1}{R}$$

as $V \rightarrow$ constant

So, B_2 will be brighter bulb Ans

Maximum Power Theorem :



Heat Produced / Power transferred to external resistor is

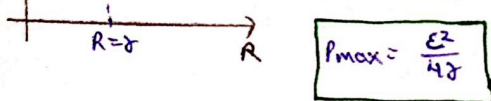
$$P_m = I^2 R = \frac{E^2 R}{(R+r)^2}$$

For P_m to be maximum,

$$\frac{dP_m}{dR} = 0 \Rightarrow R = r$$

Means, Heat produced in ext. resistor is max, if it is equal to battery's resistance

$\rightarrow$ If ext. Resistance R is less or more than r , then in both cases P_m is less than P_{max}

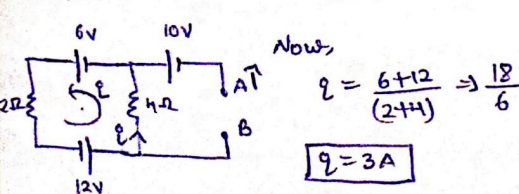
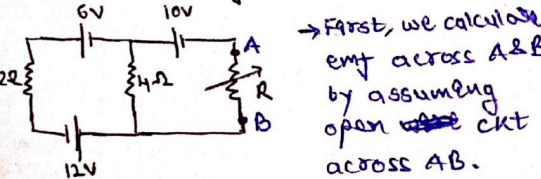


$$P_{max} = \frac{E^2}{4r}$$

Thévenin's Analysis :

$\rightarrow$ Used for making equivalent battery in any kind of circuit.

Ex: Find value of R such that it has max. power & also find P_{max} .



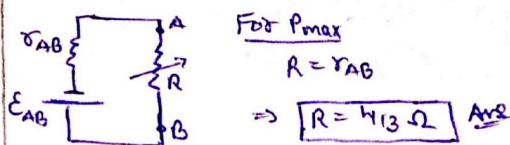
KVL eqn for open ckt loop

$$V_A + 10 + (4 \times 3) = V_B \Rightarrow V_B - V_A = 22V$$

Across AB, 2Ω & 4Ω are in parallel, so

Current Electricity-3

So, we have calculated emf across AB & internal resistance across AB. Hence entire ckt can be redrawn as

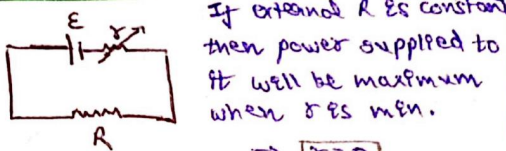


$$\Rightarrow P_{max} = 90.75W$$

Ans

$$P_{max} = \frac{E_{AB}^2}{4r} = \frac{22 \times 22 \times 3}{4 \times 4}$$

Power transfer in case of variable internal Resistance :

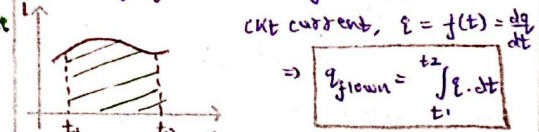


$$\Rightarrow r = 0$$

$$So, P_{max} = \frac{E^2}{R}$$

Charge flown through a circuit :

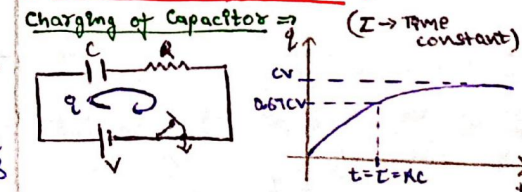
$\rightarrow$ Area enclosed b/w $i-t$ graph gives charge flown through ckt.



Heat produced in resistance due to varying current :

$$H = \int_{t_1}^{t_2} i^2 \cdot R \cdot dt$$

Transient RC - circuit :



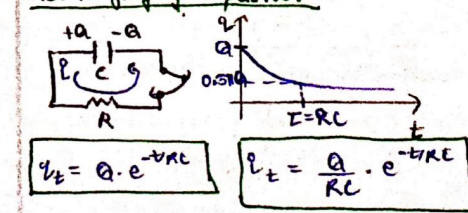
If value of $R \uparrow \Rightarrow$ slope of graph $\downarrow$
 value of $R \downarrow \Rightarrow$ slope of graph $\uparrow$

$$q_t = CV(1 - e^{-t/RC})$$

$$V = \frac{1}{C} CV^2(1 - e^{-t/RC})^2$$

$$P_m = \frac{V^2}{R} \cdot e^{-t/RC}$$

Discharging of Capacitor :



$$V_t = \frac{Q}{C} \cdot e^{-t/RC}$$

Energy stored in Capacitor

$$U_t = \frac{Q^2}{2C} \cdot e^{-2t/RC}$$

$$P_m = \frac{Q^2}{RC^2} \cdot e^{-2t/RC}$$

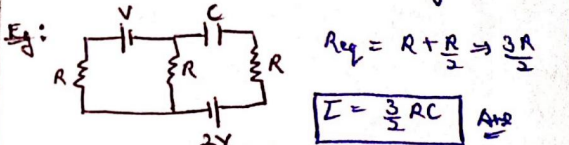
Time Constant of any RC circuit :

$$\tau = RC$$

where R is the total internal Resistance.

In case of complicated ckt's

$\rightarrow$ Replace batteries with straight wires
 $\rightarrow$ Calculate total resistance for the ckt.

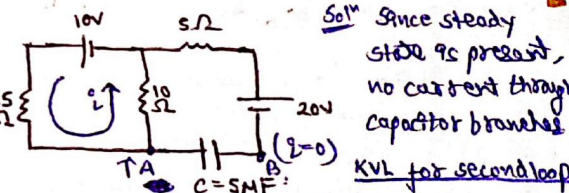


Steady State of RC circuit :

$\rightarrow$ It is the state of RC ckt when either capacitor plates have gained max. charge (while charging) or have lost max. charge (while discharging) and nothing more is possible.

$\rightarrow$ At steady state, $i = 0$ in any branch of the capacitor.

Ex: Find charge on capacitor plates at steady state.



$$i = \frac{10}{25} = 0.4A$$

$$V_A - 4 - 20 = V_B$$

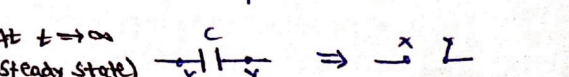
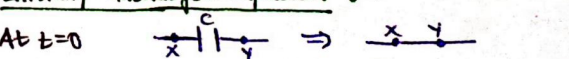
$$V_A - V_B = 24V$$

$$So, q_c = C(V_A - V_B) \Rightarrow 5 \times 24 \mu C$$

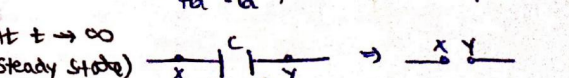
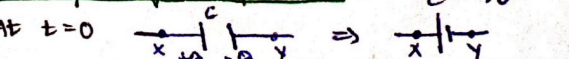
$$q_c = 120 \mu C$$
 Ans

Initial & Steady State Behaviour of Capacitors in RC ckt :

Initially Uncharged Capacitor :



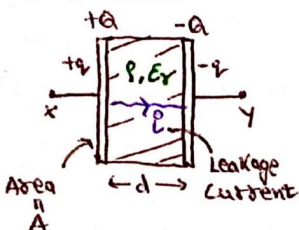
Initially Charged Capacitor :



Leaky Capacitor:

→ If the dielectric inserted b/w plates of a capacitor, ~~then~~ has some resistivity of its own,

→ Then, dielectric behaves like a resistance itself, due to which capacitor starts getting discharged and a 'Leakage current' flows through the slab.



Here,

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$R = \frac{\rho d}{A}$$

So, time constant for discharging of capacitor is

$$T = RC \Rightarrow T = \rho \epsilon_r \epsilon_0$$

Independent of dimensions of capacitor (A, d)

$$q_t = Q \cdot e^{-t/\rho \epsilon_r \epsilon_0}$$

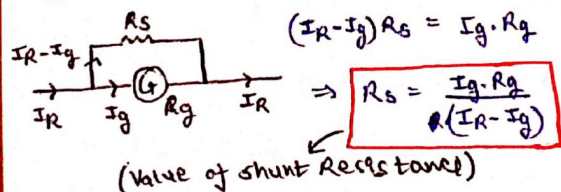
Deflection Type Meters:

Galvanometer Ammeter Voltmeter

→ Galvanometer is used to measure very small currents.
→ It has 2 parameters; R_g → Coil Resistance (0-10 Ω)
 I_g → Full Deflection Current (in mA)

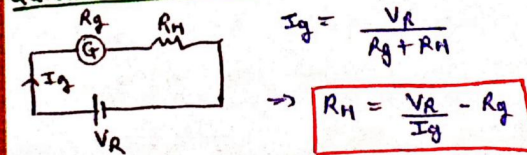
Galvanometer to Ammeter:

→ Ammeter can measure much larger currents than Galvanometer.
→ Ammeter also tells about dirn of current as it is an unidirectional instrument.



(Value of shunt Resistance)

Galvanometer to Voltmeter:

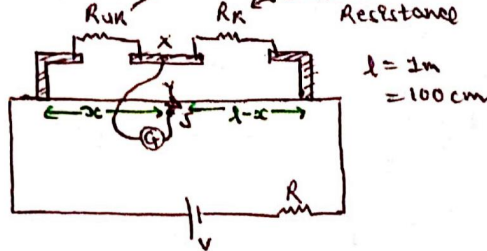


$$R_H = \frac{V_R}{I_g} - R_g$$

→ Generally, voltages are not measured by voltmeter because it has error in the readings.

Meter Bridge:

→ This instrument is used to find unknown resistance by null deflection method.



At Null deflection

$$\frac{R_{unk}}{x} = \frac{R_K}{l-x}$$

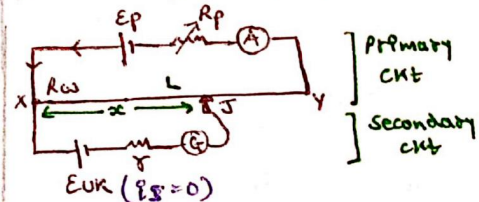
$$\Rightarrow R_{unk} = x \left(\frac{R_K}{l-x} \right)$$

Potentiometer:

→ This instrument is used to measure

- (i) Emf of unknown cells
- (ii) Internal Resistance of cells.

→ Accuracy of potentiometer is very high than voltmeter, because $\epsilon = 0$ at the point of measurement of potential difference/emf.



Process to find unknown emf:

Step-01: Calculate primary ckt current by $I_P = \frac{E_P}{R_w + R_P}$ (R_w → Wire Resistance)

Step-02: Calculate potential difference across the wire $V_{xy} = I_P \cdot R_w$

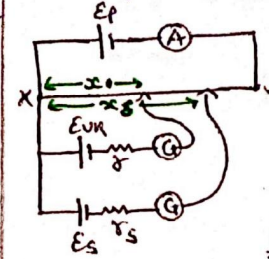
Step-03: Find potential gradient for the wire of primary ckt $\lambda = \frac{V_{xy}}{L}$

Step-04: Now, at null deflection, we simply get $E_{unk} = \lambda x$

Emf calculation by standard cell:

→ Sometimes parameters of primary ckt and potential gradient of wire are unknown.

→ In such case, we calculate E_{unk} in terms of known standard cell's emf.

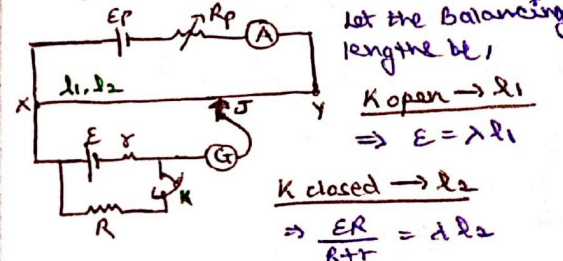


If x_s & x_u are balancing lengths for cells E_{unk} & E_s respectively, then

$$E_s = \lambda \cdot x_s \Rightarrow E_{unk} = \left(\frac{E_s}{x_s} \right) x_u$$

Internal Resistance Measurement:

→ Here, we connect a known resistance in parallel to the cell with a key.



Let the balancing length be l_1
K open $\rightarrow l_1$
 $\Rightarrow E = \lambda l_1$

K closed $\rightarrow l_2$
 $\Rightarrow \frac{E R}{R+r} = \lambda l_2$

$$\Rightarrow r = \left(\frac{l_1 - l_2}{l_2} \right) R$$

Figure of Merit:

For a galvanometer, the max. deflection current I_g is given by

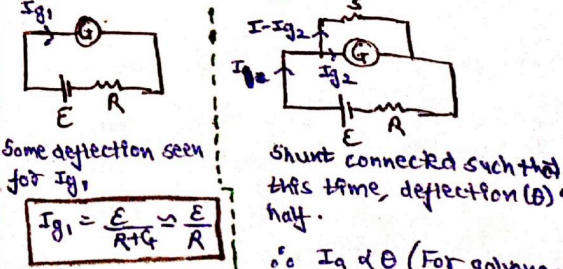
$$I_g = K n \Rightarrow \text{Max. no. of deflections.}$$

Figure of merit (const)

Half Deflection Method:

→ It is an experiment performed to find resistance of Galvanometer (G).

→ We connect large resistance with Galvanometer such that $(R \gg G)$.



Some deflection seen for I_{g1}

$$I_{g1} = \frac{E}{R+G} \approx \frac{E}{R}$$

Shunt connected such that this time, deflection (B) half.

$\therefore I_g \propto \theta$ (For galvanometer)

$$I_{g2} \cdot G = (I - I_{g2}) S \Rightarrow I_{g2} = \frac{I_{g1}}{2}$$

$$I_{g2} = \frac{I \cdot S}{S+G} \Rightarrow \frac{I_{g2}}{I_{g1}} = \frac{E}{R} \cdot \left(\frac{S}{S+G} \right)$$

$$\Rightarrow \frac{E}{R} \cdot \frac{1}{2} = \frac{E}{R} \left(\frac{S}{S+G} \right) \Rightarrow \frac{1}{2} = \frac{S}{S+G}$$

$$\Rightarrow \frac{1}{n} = \frac{S}{S+G}$$